

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397743
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Elangovan; Vivekanandh et al.

Systems and methods for detecting/preventing vehicle parts theft

Abstract

Systems and methods for implementing protections against catalytic converter theft. A vehicle may comprise Ultra-Wideband (UWB) anchors or sensors. The UWB sensors of a vehicle may be used to detect the relative distance and/or location of UWB-capable devices in close proximity to the vehicle. An assessment may be performed using the UWB anchors and/or other sensors to determine whether the presence of the unknown device is indicative of a threat to the vehicle. If a threat is detected, one or more protection steps may be taken.

Inventors: Elangovan; Vivekanandh (Canton, MI), Van Wiemeersch; John Robert (Novi, MI), Buttolo; Pietro (Dearborn Heights, MI)

Applicant: Ford Global Technologies, LLC (Dearborn, MI)

Family ID: 1000008781861

Appl. No.: 18/182213

Filed: March 10, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240300444 A1	Sep. 12, 2024

Publication Classification

Int. Cl.: B60R25/10 (20130101); B60R25/24 (20130101); B60R25/30 (20130101)

U.S. Cl.:

CPC B60R25/10 (20130101); B60R25/24 (20130101); B60R25/305 (20130101); B60R2025/1013 (20130101)

Field of Classification Search

CPC: B60R (25/10); B60R (25/24); B60R (25/305); B60R (2025/1013); G01S (5/02); G01S (13/0209); G01S (13/00); G01S (13/02); G01S (13/46); G01S (13/48); G01S (7/032); G01S (7/288); G01S (5/06); G01S (5/0257); G01S (19/12); G01S (19/256); G01S (19/48); G01S (19/49); G01S (5/0226); G01S (5/02585); G01S (5/0284); G01S (5/10); H01Q (1/3233); H01Q (1/36); H04L (63/0442)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
8963699	12/2014	Potter	N/A	N/A
9493085	12/2015	Van Wiemeersch et al.	N/A	N/A
2015/0348417	12/2014	Ignaczak	340/435	G08B 25/08
2024/0255605	12/2023	Sanji	N/A	G01S 5/0295

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
111629349	12/2019	CN	B60R 25/102
117335828	12/2023	CN	N/A
4095818	12/2021	EP	N/A
2020030895	12/2019	JP	N/A
102297848	12/2017	KR	H04L 9/40
102109695	12/2019	KR	N/A
20220091981	12/2021	KR	N/A

OTHER PUBLICATIONS

Intermotive Introduces Anti-Catalytic Converter Theft Alarm, News/Media Release, Nov. 15, 2022, 1-6. cited by applicant

Primary Examiner: Zimmerman; Brian A

Assistant Examiner: Afrifa-Kyei; Anthony D

Attorney, Agent or Firm: Eversheds Sutherland (US) LLP

Background/Summary

BACKGROUND

(1) There is a growing need for enhanced detection/protection of catalytic converters from theft. While there are a variety of techniques for detecting theft of vehicle parts such as catalytic converters, implementing new hardware-based protections can be expensive, and can present

challenges to owners of existing cars, as retrofitting existing vehicles will take time, and not all vehicle owners may have the time, availability, or budget for having new hardware installed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A detailed description is set forth regarding the accompanying drawings. The use of the same reference numerals may indicate similar or identical items. Various embodiments may utilize elements and/or components other than those illustrated in the drawings, and some elements and/or components may not be present in various embodiments. Elements and/or components in the figures are not necessarily drawn to scale. Throughout this disclosure, depending on the context, singular and plural terminology may be used interchangeably.

(2) FIG. 1 illustrates an example vehicle in accordance with the teachings herein.

(3) FIG. 2 is a block diagram of electronic components of the vehicle of FIG. 1.

(4) FIG. 3 depicts a diagram in which catalytic converter theft can be proactively detected, according to at least one embodiment of the present disclosure.

(5) FIG. 4 shows an illustrative example of a process for detecting and/or preventing catalytic converter theft, in accordance with one or more example embodiments of the present disclosure.

DETAILED DESCRIPTION

Overview

(6) The present disclosure is directed to systems and methods for detecting and/or preventing the theft of components of vehicles, such as catalytic converter theft. There is a growing need for enhanced detection/protection of catalytic converters from theft. While there are a variety of techniques for detecting theft of vehicle parts such as catalytic converters, implementing new hardware-based protections can be expensive, and can present challenges to owners of existing cars, as retrofitting existing vehicles will take time, and not all vehicle owners may have the time, availability, or budget for having new hardware installed. Accordingly, while the illustrative embodiment contemplates theft of a catalytic converter, this invention could apply to any element or component attached to a vehicle underbody, such as a battery, suspension component, wiring, etc.

(7) Techniques described herein contemplate, in at least some embodiments, a software-based solution for protecting against catalytic converter theft that involves the use of a vehicle's existing Ultra-Wideband (UWB) controller and anchors and electronic systems to detect potential catalytic converter theft. An additional advantage of the solutions presented herein is that these techniques can potentially detect attempted theft earlier than other systems, possibly avoiding any damage to the catalyst, by proactively waking sensors before malicious actors have even touched the vehicle.

(8) Various embodiments of the present disclosure may be implemented in the context of a vehicle. A vehicle may comprise a plurality of Ultra-Wideband (UWB) primary transceivers and/or anchors located throughout the vehicles interior and/or exterior, here forward to all be referred to as anchors. The vehicle may comprise electronic systems that include one or more processors and memory storing executable instructions that, as a result of execution by the one or more processors, causes the vehicle to perform various processes described throughout this disclosure (e.g., processes in accordance with FIG. 4). In at least one embodiment, the vehicle will determine, using at least one of the UWB anchors, presence of an unknown device; determine, using the at least one UWB anchor, a distance of the unknown device to the vehicle; determine, based on the distance, that the unknown device has crossed a proximity threshold of the vehicle; determine, using the plurality of UWB anchors, a location of the unknown device; determine, based on the location of the unknown device, that the unknown device has breached the vehicle's environment; and based at least in part on the determination that the unknown device has breached the vehicle's environment,

perform one or more protection steps.

(9) In at least one embodiment, the vehicle will further: responsive to the determination that the unknown device has breached the vehicle's environment, activate one or more sensors of the vehicle, based at least in part on the location of the unknown device; determine, based on additional data collected by the one or more sensors of the vehicle, a threat level; and determine the one or more protection steps based at least in part on the threat level.

(10) In at least one or more embodiments, the vehicle comprises audio and visual sensors that can be used to assess threats to the vehicle. In at least one embodiment, the vehicle comprises a camera underneath the vehicle that can be activated to determine whether there is a threat of catalytic converter theft.

(11) In various embodiments, the vehicle can perform a series of escalating protection steps that includes activation of a first alarm (e.g., one chirp) and activation of a second alarm (e.g., three chirps), responsive to a determination that the unknown device is still within the proximity threshold after the activation of the first alarm.

(12) In various embodiments, the proximity threshold is determined based at least in part on time of day or other situational factors such as whether the vehicle is located on a busy street.

(13) In various embodiments, the vehicle will determine that a known device associated with the vehicle is not within proximity to the vehicle as part of its threat assessment. The proximity of the known device may be different from the threshold proximity for unknown devices.

(14) Turning to the figures, FIG. 1 illustrates an example vehicle **100** in accordance with the teachings herein. The vehicle **100** may be a standard gasoline powered vehicle, a hybrid vehicle, an electric vehicle, a fuel cell vehicle, and/or any other mobility implement type of vehicle. The vehicle **100** includes parts related to mobility, such as a powertrain with an engine, a transmission, a suspension, a driveshaft, and/or wheels, etc. The vehicle **100** may be non-autonomous, semi-autonomous (e.g., some routine motive functions controlled by the vehicle **100**), or autonomous (e.g., motive functions are controlled by the vehicle **100** without direct driver input).

(15) In the illustrated example, the vehicle **100** includes tires **102** and an engine **104**. For example, each of the tires **102** is coupled to a body of the vehicle **100** at a different respective wheel well of the vehicle **100** to enable the vehicle **100** to travel along a road surface. The engine **104** of the illustrated example is an internal combustion engine, an electric motor, and/or any other power source that propels movement of the vehicle **100**. In some examples, the engine **104** is initially activated upon receiving energy from a starter battery and subsequently is powered via energy receiving from an alternator.

(16) The vehicle **100** of the illustrated example also includes a cabin **106**, a front side **108**, a rear side **110**, a driver side **112**, and a passenger side **114**. The front side **108** includes a front windshield **116** and a front panel **118** (also referred to as a hood), and the rear side **110** includes a rear windshield **120** and a rear panel **122** (also referred to as a tailgate panel and/or a trunk panel). The vehicle **100** includes doors **124** that include side windows **126** and door panels **128**. For example, the driver side **112** of the vehicle **100** includes one of the doors **124** (e.g., a driver-side door) that includes one of the side windows **126** and one of the door panels **128**. Further, the passenger side **114** includes one of the doors **124** (e.g., a passenger-side door) that includes another of the side windows **126** and another of the door panels **128**. As illustrated in FIG. 1, the front windshield **116**, the front panel **118**, the rear windshield **120**, the rear panel **122**, the side windows **126**, and the door panels **128** form, at least in part, an outer layer of the vehicle **100**. That is, the outer layer of the vehicle **100** includes the front windshield **116**, the front panel **118**, the rear windshield **120**, the rear panel **122**, the side windows **126**, and the door panels **128**. Additionally, the cabin **106** is defined, at least in part, by the front windshield **116**, the rear windshield **120**, and the doors **124**. In some examples, the front windshield **116** is formed of laminated glass, and the rear windshield **120** and the side windows **126** are formed of non-laminated, tempered glass. Further, the front panel **118**, the rear panel **122**, and the door panels **128** are formed of, for example, steel, aluminum, other

metallic material, plastic, carbon fiber, fiberglass, other composite material, and/or any combination thereof.

(17) As illustrated in FIG. 1, the vehicle **100** includes one or more communication nodes **130**. In the illustrated example, the communication nodes **130** are configured to wirelessly detect the presence of UWB-capable devices such as device **132** of an unknown person **134**. In various embodiments, device **132** emits UWB (Ultra-Wideband) wireless signals and one or more anchors or sensors of vehicle **100** are subscribers for such signals and can use the signal to determine the relative distance or location of device **132**. The location of device **132** may be used to infer the location of unknown person **134** relative to vehicle **100**. In various embodiments, the unknown person **134** carries a mobile device with UWB emitters that are configured by default to periodically emit UWB wireless communications which can be received by the communication nodes **130**. Device **132** may refer to any suitable device that emits UWB signals at regular intervals.

(18) As noted above, while various embodiments described in connection with FIG. 1 discuss the use of UWB anchors, these are merely non-limiting examples of embodiments that are contemplated within the scope of this disclosure. In various embodiments, a combination of UWB transceivers and/or anchors are utilized to determine relative position and/or distance of UWB-capable devices near the vehicle.

(19) The communication nodes **130** include hardware and firmware to establish a wireless connection with the device **132**. In various embodiments, the wireless connection between the device **132** and the communication nodes **130** is a two-way communications channel that is implemented according to a specification, for example, according to an IEEE or CCC defined format. In various embodiments, the standard will be automated in the driver layer of the respective devices, such as the vehicle **100** and/or device **132** depicted in FIG. 1. As an example, the vehicle **100** and/or device **132** may send a round-trip message to the other as part of UWB Time-of-Flight (ToF) ranging that can be used to estimate the distance between the vehicle **100** and device **132**. Vehicle **100** may utilize UWB signals emitted by the device **132** to determine the relative distance of the device **132**, perform localization to determine the relative location of the device (e.g., whether the device is located behind the vehicle, in front of vehicle, at an angle, and so on), and implement various protective steps based on the threat level that is detected. For example, in this sense, the device **132** may be viewed as a data producer and the communication nodes **130** as data consumers. While two-way communications are described above, there may be other embodiments in which one-way communications are possible.

(20) In various embodiments, the communication nodes **130** are short-range wireless modules that wirelessly communicate with nearby devices such as the device **132** of the unknown person **134** via short-range wireless communication protocols. In some examples, the communication nodes **130** implement the Bluetooth® and/or Bluetooth® Low Energy (BLE) protocols. The Bluetooth® and BLE protocols are set forth in Volume 6 of the Bluetooth® Specification 4.0 (and subsequent revisions) maintained by the Bluetooth® Special Interest Group. In other examples, the communication nodes **130** may use WiFi, WiMax, NFC, UWB (Ultra-Wideband), and/or any other communication protocol that enables the communication nodes **130** to receive data from the device **132**.

(21) In various embodiments, device **132** emits UWB signals to nearby subscribers (e.g., communication nodes **130**) without requiring the subscribers to perform a pairing or identification process. Accordingly, it should be appreciated that vehicle **100** may receive UWB signals from device **132** even when unknown person **134** is “unknown” from the perspective of vehicle **100**—for example, unknown person **134** may be a stranger that is walking by vehicle **100** for the first time and has never previously interacted (e.g., either physically or electronically) with vehicle **100**. In various embodiments, communication nodes **130** are able to detect the presence of device **132** simply by virtue of the device **132** being in physical proximity with the vehicle **100** and not

requiring any additional configuration, setup, or application settings by unknown person **134**. (22) Further, the vehicle **100** of the illustrated example includes a communication module **136** that includes wired or wireless network interfaces to enable communication with external networks (e.g., a network **216** of FIG. 2). The communication module **136** also includes hardware (e.g., processors, memory, storage, antenna, etc.) and software to control the wired or wireless network interfaces. In the illustrated example, the communication module **136** includes one or more communication controllers for standards-based networks (e.g., Global System for Mobile Communications (GSM), Universal Mobile Telecommunications System (UMTS), Long Term Evolution (LTE), Code Division Multiple Access (CDMA), WiMAX (IEEE 802.16m); Near Field Communication (NFC); local area wireless network (including IEEE 802.11 a/b/g/n/ac or others), dedicated short range communication (DSRC), and Wireless Gigabit (IEEE 802.11ad), etc.). In some examples, the communication module **136** includes a wired or wireless interface (e.g., an auxiliary port, a Universal Serial Bus (USB) port, a Bluetooth® wireless node, etc.) to communicatively couple with a mobile device (e.g., a smart phone, a wearable, a smart watch, a tablet, etc.). In such examples, the vehicle **100** may communicate with the external network via the coupled mobile device. The external network(s) may be a public network, such as the Internet; a private network, such as an intranet; or combinations thereof, and may utilize a variety of networking protocols now available or later developed including, but not limited to, TCP/IP-based networking protocols. In various embodiments, communication module **136** is configured to communicate with the vehicle's owner (e.g., a person other than unknown person **134**) via a paired device. For example, if unknown person **134** attempts to slide under vehicle **100**, UWB anchors may be used to triangulate the location of device **132** and wake up or activate additional sensors (e.g., audio and/or visual sensors) to determine whether an unauthorized activity (e.g., catalytic converter **142** theft) is occurring. Additionally, if such an activity is detected, alarms may be activated and/or the owner of vehicle **100** will be notified via communication module **136** as to the activity. Catalytic converter **142** may be located underneath the vehicle and is depicted in FIG. 1 solely for exposition purposes only.

(23) While a single catalytic converter is depicted in FIG. 1, it is noted that other embodiments contemplated within the scope of this disclosure include vehicles that have an arrangement of multiple catalytic converters. Various techniques described herein may be applied to protect against threats to some or all of the catalytic converters of such a vehicle, including the ability to simultaneously monitor multiple components at the same time. For example, on a vehicle with two catalytic converters, the vehicle's sensors may be configured to monitor and protect both catalytic converters from theft and determine a threat level to each when unknown devices are detected near the vehicle.

(24) The vehicle **100** also includes vibration sensors **138** that are coupled to the outer layer of the vehicle **100** to detect audio vibrations. For example, each of the vibration sensors **138** are rigidly mounted to interior surface(s) and/or exterior surface(s) of the outer layer of the vehicle **100**. The vibration sensors **138** may be coupled to the interior surface(s) of the outer layer of the vehicle **100** to prevent the vibration sensors **138** from being contacted by other objects. In other examples, one or more of the vibration sensors **138** are embedded within the outer layer of the vehicle **100**.

(25) Further, in the illustrated example, each side of the vehicle **100** includes one of the vibration sensors **138**. That is, one of the vibration sensors **138** is coupled to each of the front side **108**, the rear side **110**, the driver side **112**, and the passenger side **114** of the vehicle **100**. In other examples, more or less of the vibration sensors **138** may be coupled to one or more of the front side **108**, the rear side **110**, the driver side **112**, and/or the passenger side **114** of the vehicle **100**. Further, in the illustrated example, one of the vibration sensors **138** is coupled to each of the front windshield **116**, the rear windshield **120**, and the side windows **126**. In other examples, more or less of the vibration sensors **138** may be coupled to one or more of the front windshield **116**, the rear windshield **120**, and/or one or more of the side windows **126**. Further, in some examples, one or more of the

vibration sensors **138** is coupled to the front panel **118**, the rear panel **122**, and/or one or more of the door panels **128**.

(26) The vibration sensors **138** of the illustrated example are configured to measure audio vibrations of portions of the outer layer of the vehicle **100** to which the vibration sensors **138** are coupled. The audio vibrations are caused by sound waves that are impinging upon the outer layer. For example, sound waves travelling through a medium (e.g., the front windshield **116**, the front panel **118**, the rear windshield **120**, the rear panel **122**, one or more of the side windows **126**, one or more of the door panels **128**) cause the medium to oscillate and/or vibrate.

(27) The vibration sensors **138** measure these audio vibrations caused by the sound waves when the sound waves impinge upon the corresponding portions of the outer layer of the vehicle **100** and generate electrical signals that correspond to the measured audio vibrations. The electrical signals may be subsequently processed and/or analyzed to determine the sound waves that impinged upon the outer layer of the vehicle **100**. For example, the vibration sensors **138** are accelerometers (e.g., uni-axial accelerometers, tri-axial accelerometers, micro-machined accelerometers, piezoelectric accelerometers, etc.), piezoelectric contact films, microphones, and/or any other devices that are capable of detecting sound by monitoring audio vibrations of the outer layer. In the illustrated example, the vibration sensors **138** are configured to measure audio vibrations that are perpendicular to surface(s) to which the vibration sensors **138** are coupled.

(28) In other words, the outer layer of the vehicle **100** and the vibration sensors **138** form microphones. For example, the portions of the outer layer that couple to the vibration sensors **138** function as diaphragms of the microphones. In the illustrated example, the front windshield **116** and one of the vibration sensors **138** that is coupled to the front windshield **116** form one microphone, the rear windshield **120** and one of the vibration sensors **138** that is coupled to the rear windshield **120** form another microphone, and the side windows **126** of those of the vibration sensors **138** that are coupled to the side windows **126** form other microphones. In other examples, the front panel **118** forms a microphone when one of the vibration sensors **138** is coupled to the front panel **118**, the rear panel **122** forms a microphone when one of the vibration sensors **138** is coupled to the rear panel **122**, and/or one or more of the door panels **128** forms a microphone with a corresponding one or more of the vibration sensors **138**.

(29) As illustrated in FIG. 1, the vehicle **100** also includes audio actuators **140** (also referred to as vibration speaker audio actuators) that are coupled to the outer layer of the vehicle **100** to vibrate the outer layer. For example, each of the audio actuators **140** (e.g., of SoundBug devices) are rigidly mounted to interior surface(s) and/or exterior surface(s) of the outer layer of the vehicle **100**. The audio actuators **140** may be coupled to the interior surface(s) of the outer layer of the vehicle **100** to prevent the audio actuators **140** from being contacted by other objects. In other examples, one or more of the audio actuators **140** are embedded within the outer layer of the vehicle **100**.

(30) Further, in the illustrated example, each side of the vehicle **100** includes one of the audio actuators **140**. That is, one of the audio actuators **140** is coupled to each of the front side **108**, the rear side **110**, the driver side **112**, and the passenger side **114** of the vehicle **100**. In other examples, more or less of the audio actuators **140** may be coupled to one or more of the front side **108**, the rear side **110**, the driver side **112**, and/or the passenger side **114** of the vehicle **100**. Further, in the illustrated example, one of the audio actuators **140** is coupled to each of the front windshield **116**, the rear windshield **120**, and the side windows **126**. In other examples, more or less of the audio actuators **140** may be coupled to one or more of the front windshield **116**, the rear windshield **120**, and/or one or more of the side windows **126**. Further, in some examples, one or more of the audio actuators **140** is coupled to the front panel **118**, the rear panel **122**, and/or one or more of the door panels **128**.

(31) The audio actuators **140** of the illustrated example are configured to emit sound by vibrating corresponding portions of the outer layer of the vehicle **100** to which the audio actuators **140** are

coupled. For example, each of the audio actuators **140** includes an arm that actuates upon receiving an electrical signal. For each of the audio actuators **140**, the arm contacts or causes another actuator component to contact an adjacent surface of the outer layer as the arm actuates to cause the outer layer to vibrate. That is, each of the audio actuators **140** functions as an adjustable tuning fork by actuating an adjacent portion of the outer layer to cause that portion of the outer layer to form an acoustic baffle of a speaker. In other words, the outer layer of the vehicle **100** and the audio actuators **140** form speakers. In the illustrated example, the front windshield **116** and one of the audio actuators **140** that is coupled to the front windshield **116** form one speaker, the rear windshield **120** and one of the audio actuators **140** that is coupled to the rear windshield **120** forms another speaker, and the side windows **126** of those of the audio actuators **140** that are coupled to the side windows **126** form other speakers. In other examples, the front panel **118** forms a speaker when one of the audio actuators **140** is coupled to the front panel **118**, the rear panel **122** forms a speaker when one of the audio actuators **140** is coupled to the rear panel **122**, and one or more of the door panels **128** form a speaker with a corresponding one or more of the audio actuators **140**.

(32) In various embodiments, vehicle **100** comprises additional cameras not depicted in FIG. **1**. For example, video cameras (not depicted in FIG. **1**) may be located on each side of vehicle **100** and may provide 360-degrees Coverage of the vehicle **100**. The cameras or other visual sensors may be in communication with various other components of vehicle **100** such as communication module **136**. In various embodiments, one or more visual sensors are located underneath vehicle **100** that can be used to detect the presence of unauthorized individuals underneath the vehicle. In various embodiments, a video feed can be provided by vehicle **100** to a remote server using communication module **136** to process the video feed and perform various techniques such as object detection and/or object recognition.

(33) FIG. **2** is a block diagram of electronic components **200** of the vehicle **100**. In the illustrated example, the electronic components **200** include an on-board computing platform **202**, the communication nodes **130**, the communication module **136**, the audio actuators **140**, sensors **204**, electronic control units (ECUs) **206**, and a vehicle data bus **208**.

(34) The on-board computing platform **202** includes a microcontroller unit, controller or processor **210**; memory **212**; and a database **214**. In some examples, the processor **210** of the on-board computing platform **202** is configured to implement some or all of the functionality described below in connection with FIG. **4**.

(35) The processor **210** may be any suitable processing device or set of processing devices such as, but not limited to, a microprocessor, a microcontroller-based platform, an integrated circuit, one or more field programmable gate arrays (FPGAs), and/or one or more application-specific integrated circuits (ASICs). The memory **212** may be volatile memory (e.g., RAM including non-volatile RAM, magnetic RAM, ferroelectric RAM, etc.), non-volatile memory (e.g., disk memory, FLASH memory, EPROMs, EEPROMs, memristor-based non-volatile solid-state memory, etc.), unalterable memory (e.g., EPROMs), read-only memory, and/or high-capacity storage devices (e.g., hard drives, solid state drives, etc.). In some examples, the memory **212** includes multiple kinds of memory, particularly volatile memory and non-volatile memory.

(36) The memory **212** is computer readable media on which one or more sets of instructions, such as the software for operating the methods of the present disclosure, can be embedded. The instructions may embody one or more of the methods or logic as described herein. For example, the instructions reside completely, or at least partially, within any one or more of the memory **212**, the computer readable medium, and/or within the processor **210** during execution of the instructions.

(37) The terms “non-transitory computer-readable medium” and “computer-readable medium” include a single medium or multiple media, such as a centralized or distributed database, and/or associated caches and servers that store one or more sets of instructions. Further, the terms “non-transitory computer-readable medium” and “computer-readable medium” include any tangible medium that is capable of storing, encoding or carrying a set of instructions for execution by a

processor or that cause a system to perform any one or more of the methods or operations disclosed herein. As used herein, the term “computer readable medium” is expressly defined to include any type of computer readable storage device and/or storage disk and to exclude propagating signals. (38) Further, in the illustrated example, the communication module **136** of the vehicle **100** is in wireless communication with a network **216**. For example, the communication module **136** is communicatively coupled to the network **216**.

(39) The sensors **204** are arranged in and around the vehicle **100** to monitor properties of the vehicle **100** and/or an environment in which the vehicle **100** is located. One or more of the sensors **204** may be mounted to measure properties around an exterior of the vehicle **100**. Additionally or alternatively, one or more of the sensors **204** may be mounted inside a cabin of the vehicle **100** or in a body of the vehicle **100** (e.g., an engine compartment, wheel wells, etc.) to measure properties in an interior of the vehicle **100**. For example, the sensors **204** include accelerometers, odometers, tachometers, pitch and yaw sensors, wheel speed sensors, microphones, tire pressure sensors, biometric sensors and/or sensors of any other suitable type. In the illustrated example, the sensors **204** include the vibration sensors **138** which may act as microphones for collecting audio data.

(40) The ECUs **206** monitor and control the subsystems of the vehicle **100**. For example, the ECUs **206** are discrete sets of electronics that include their own circuit(s) (e.g., integrated circuits, microprocessors, memory, storage, etc.) and firmware, sensors, actuators, and/or mounting hardware. The ECUs **206** communicate and exchange information via a vehicle data bus (e.g., the vehicle data bus **208**). Additionally, the ECUs **206** may communicate properties (e.g., status of the ECUs **206**, sensor readings, control state, error and diagnostic codes, etc.) to and/or receive requests from each other. For example, the vehicle **100** may have seventy or more of the ECUs **206** that are positioned in various locations around the vehicle **100** and are communicatively coupled by the vehicle data bus **208**.

(41) In the illustrated example, the ECUs **206** include an engine control unit **218** and a body control module **220**. For example, the engine control unit **218** control(s) operation (e.g., engine idle shutdown timer) of the engine **104** of the vehicle **100**. Further, the body control module **220** controls one or more subsystems throughout the vehicle **100**, such as power windows, power locks, an immobilizer system, power mirrors, etc. For example, the body control module **220** includes circuits that drive one or more of relays (e.g., to control wiper fluid, etc.), brushed direct current (DC) motors (e.g., to control power seats, power locks, power windows, wipers, etc.), stepper motors, LEDs, etc.

(42) The vehicle data bus **208** communicatively couples the communication nodes **130**, the communication module **136**, the audio actuators **140**, the on-board computing platform **202**, the sensors **204**, and the ECUs **206**. In some examples, the vehicle data bus **208** includes one or more data buses. The vehicle data bus **208** may be implemented in accordance with a controller area network (CAN) bus protocol as defined by International Standards Organization (ISO) 11898-1, a Media Oriented Systems Transport (MOST) bus protocol, a CAN flexible data (CAN-FD) bus protocol (ISO 11898-7) and/a K-line bus protocol (ISO 9141 and ISO 14230-1), and/or an Ethernet™ bus protocol IEEE 802.3 (2002 onwards), etc.

(43) FIG. 3 depicts a diagram **300** in which catalytic converter theft can be proactively detected, according to at least one embodiment of the present disclosure.

(44) Turning to FIG. 3, vehicle **302** may be a vehicle that comprises a catalytic converter **304** and various sensors suites, such as those described in connection with FIG. 1 and/or FIG. 2.

(45) In various embodiments, vehicle **302** comprises several Ultra-Wideband (UWB) anchors **312** or sensors located on the interior and/or exterior of the vehicle that can be used to detect UWB signals that are emitted by devices (e.g., devices **306**, **308**, and **310**) and use triangulation techniques to resolve the location of the emitter device. For example, in various embodiments, the device location is determined via UWB using Time-of-Flight (ToF) data between a device and multiple anchors and performing triangulation on the ToF results from the multiple anchors. While

various embodiments described in connection with FIG. 3 discuss the use of UWB anchors, these are merely non-limiting examples of embodiments that are contemplated within the scope of this disclosure. In various embodiments, a combination of UWB transceivers and/or anchors are utilized to determine relative position and/or distance of UWB-capable devices near the vehicle.

(46) Vehicle **302** may have installed on its electronic systems software for detecting potential catalytic converter theft. The software may be in the form of application code that interfaces with various sensors located across the device. This software, in various embodiments, will continuously detect any nearby UWB signals and determine how far away a device is from the vehicle **302**.

(47) In various embodiments, vehicle **302** is able to distinguish between known devices and unknown devices. It may be implied that a known device is associated with a known user and an unknown device is associated with an unknown user. A known device, in various embodiments, refers to a device that has been paired with vehicle **302** or is otherwise associated with the vehicle **302** in some way. Examples of known devices of the vehicle's owner include the vehicle's "key fob", or an electronic device that wirelessly communicates with the vehicle, or a "phone-as-a-key" that refers to a mobile device (e.g., a smart phone, a wearable, a smart watch, a tablet, etc.) that includes hardware and/or software to function as a key fob, and so on and so forth. Conversely, an unknown device, in various embodiments, refers to a device that has not been previously paired or otherwise associated with a vehicle.

(48) FIG. 3 depicts a vehicle **302** and three unknown devices—device **306**, device **308**, and device **310**, that may be owned by malicious users seeking to steal a catalytic converter **304**.

(49) In various embodiments, devices **306-310** are devices that conform to the Car Connectivity Consortium (CCC) standard for Ultra-Wideband based vehicle key solutions. In various embodiments, as part of implementing the CCC standard, the devices emit UWB signals on a regular basis and allow for communications (e.g., two-way) communications between a chip inside the device and another UWB device, such as a vehicle's UWB anchors **312**. It should be noted that, in various embodiments, there is no requirement for the device and UWB anchor to be paired or otherwise perform a formal handshake in order for the UWB anchors to receive the device's UWB signals.

(50) In various embodiments, the devices **306-310** are configured to always advertise their position via UWB signals via an open application programming interface (API). This may be in the form of a "ping" in which the device (e.g., devices **306-310**) broadcast their position to the vehicle's UWB anchors to allow for the location of the device to be ascertained. Devices that implement such technology may include, without limitation, smartphones, smart watches, fitness trackers, tablets, or other types of portable electronic devices.

(51) It should be noted that according to various implementations, there is no requirement of downloading any app by the unknown device's owners for the vehicle to be able to detect the devices (e.g., devices **306-310** owned by unknown individuals). In various embodiments, devices **306-310** are devices implemented according to one or more CCC Ultra-Wideband (UWB) standards that are natively supported by the devices' operating systems, such as iOS or Android. UWB pinging may be built into these devices so that UWB signals periodically ping to make the beacons (e.g., devices) discoverable by the UWB anchors of the vehicle.

(52) Further, in various implementations, the UWB signals do not pose a privacy concern to devices owners. Generally speaking, the UWB signals may provide only a unique identifier that can be used to determine whether a device is a known device or unknown device from the perspective of the vehicle. That is, if the standard CCC beacon is heard by the vehicle (which may be unavoidable with this technology for any receiver) and it is not a device that is registered, authorized, or otherwise known to the vehicle, then the vehicle's alarm and alert systems may be activated to evaluate whether the presence of the unknown device is indicative of a threat to the vehicle.

(53) Continuing with FIG. 3, devices (e.g., any of devices **306-310**) may be continuously emitting

plings that are detected by UWB anchors **312** of vehicle **302**. The vehicle may triangulate the position of the devices by measuring how long it takes for the UWB signals to reach the various anchors. The use of UWB frequency in the range of 3.1 to 10.6 GHz can provide an accuracy within centimeters, meaning that this type of technology is highly accurate for determining the location of the emitter devices. In various embodiments, the UWB frequency range used by the devices is in the range of 6 to 8 GHz. When a device is in proximity to vehicle **302**, the vehicle may determine the location of the device relative to the vehicle using a suite of UWB anchors.

(54) If vehicle **302** detects that there is an unknown device within the proximity threshold, then the vehicle may take a set of protective measures, which may be designed to protect the vehicle against theft or damage. The vehicle may perform localization steps to determine where the unknown device is relative to the vehicle—for example, device **310** being located to the left of the vehicle may trigger camera sensors of the left side of the vehicle to activate. Additionally, if a device is detected as being inside the perimeter of the vehicle (e.g., device **306** or device **308**) then a broad range of sensors may be activated. For example, camera or other sensors disposed of underneath the vehicle may be activated to determine whether the unknown person is underneath the car and either preparing to or in the process of catalytic converter theft. In some embodiments, alarms can be raised to attract attention and deter theft. In some embodiments, a notification is transmitted to a known user's device and may include media (e.g., a video clip) in real-time. In various embodiments, there are different levels of responses that can be applied based on a variety of contexts, which may be based on the relative location of the unknown device or contextual clues. Based on how much certainty there is of a breach, the vehicle's response systems may take an appropriate action based on assessment of the environment using additional sensors beyond the UWB sensor suite.

(55) It is noted that one goal of the system is to minimize the number of false alarms, since repeatedly activating cameras, microphones, and other “high-power” components could prematurely discharge the vehicle's battery. Accordingly, the vehicle may categorize the level of perceived threat and take an appropriate action for the detected threat level. The formula used to classify level of threat can be calculated based on proximity of device to vehicle, and how long device is detected stationary nearby vehicle.

(56) Other sensors in the vehicle that are powered in stand-by can also be used to enhance the formula. For example, capacitive sensors on door-handle and capacitive sensor on rear bumper could detect if someone is touching the side of the vehicle as they crouch and lie down to slide under. In various embodiments, piezo sensors used in “window-as-a-microphone” can be used in continuous low power mode to detect any low impact caused by someone touching the vehicle.

(57) A “positive” signal (in the sense that something was detected by the sensors) from these extra sensors could elevate an otherwise low-level threat based solely on UWB to high level threat, and vice-versa. Alternatively, presence or absence of signal on these extra sensors could be used to lower or raise the threshold value used in the threat level formula.

(58) Additionally, location of where the vehicle is parked, and the time of day could also be used to alter the thresholds. A vehicle parked in a crowded street in full daylight might require a more “robust” signal before threat is set as high. On the other hand, a vehicle in an isolated parking lot at night will turn on its cameras at the minimum sign of danger. In various embodiments, the proximity threshold depicted in FIG. 3 can be expanded or reduced in size based on such situational factors.

(59) In order to reduce false alarms, the vehicle **302** may be configured to deactivate these alarms under certain conditions. For example, when a valid key is within the cabin of the vehicle or nearby the vehicle, then the protection measures described herein may be deactivated, as the presence of a known device near the vehicle can be used as an implicit hint that theft is unlikely. In various embodiments, the protection measures described herein are deactivated if the vehicle is at certain known locations. For example, if the vehicle is located at a certified mechanic's shop or other

authorized auto shop, it may be appropriate to disable the protection mechanisms to allow for certain expected service work to be performed without triggering false alarms. In various embodiments, a known user can use a smartphone application to deactivate (e.g., temporarily) the protection mechanisms described herein for any reason.

(60) FIG. 4 shows an illustrative example of a process **400** for detecting and/or preventing catalytic converter theft, in accordance with one or more example embodiments of the present disclosure. In at least one embodiment, some or all of the process **400** (or any other processes described herein, or variations and/or combinations thereof) is performed under the control of one or more computer systems that store computer-executable instructions and may be implemented as code (e.g., computer-executable instructions, one or more computer programs, or one or more applications) executing collectively on one or more processors, by hardware, software, or combinations thereof. The code, in at least one embodiment, is stored on a computer-readable storage medium in the form of a computer program storing a plurality of computer-readable instructions executable by one or more processors. The computer-readable storage medium, in at least one embodiment, is a non-transitory computer-readable medium. In at least one embodiment, at least some of the computer-readable instructions usable to perform the process **400** are not stored solely using transitory signals (e.g., a propagating transient electric or electromagnetic transmission). A non-transitory computer-readable medium does not necessarily include non-transitory data storage circuitry (e.g., buffers, caches, and queues) within transceivers of transitory signals. Process **400** may be implemented in the context of various systems and methods described elsewhere in this disclosure, such as those discussed in connection with FIGS. 1-3. In at least one embodiment, process **400** or a portion thereof is collectively implemented by an electronic system of a vehicle such as those described in connection with FIGS. 1-3.

(61) In at least one embodiment, process **400** starts with a step **402** in which a vehicle detects an unknown device and determines the relative distance of the device to the vehicle using Ultra-Wideband (UWB) anchors of the vehicle. In various embodiments, the unknown device is a device that is not paired or otherwise associated with the vehicle. The device may be a smartphone device or smartwatch device whose operating system periodically emits UWB signals that the UWB anchors of the vehicle can use to determine the relative distance of the device to the vehicle. In various embodiments, the vehicle utilizes UWB transceivers in place of or in addition to UWB anchors to perform various steps of process **600**.

(62) Once the relative distance of the unknown device has been determined, step **404** may involve determining whether unknown device has crossed a proximity threshold of the vehicle. The proximity threshold may be any suitable shape around the vehicle and the size of the proximity threshold may change based on situational factors. For example, the proximity threshold may be smaller when the vehicle is parked on a busy street in the middle of the day, whereas the proximity threshold may be larger when the vehicle is in a remote parking lot in the middle of the night. If the detected device has not crossed the proximity threshold at step **404**, it may be considered to be of minimal or no threat to the vehicle and the process **400** may return to step **402** and continue to use UWB anchors to monitor for threats.

(63) Assuming, however, that the unknown device has crossed the proximity threshold, the vehicle may perform step **406** and localize the unknown device and determine the location of the unknown device. Triangulation techniques may be utilized so that the UWB signals emitted by the unknown device are collected by multiple UWB sensors which can be used to determine a location of the device and not just its distance. For example, through the use of the multiple UWB anchors located throughout the vehicle, the vehicle's electronic systems can determine a location of the unknown device relative to the vehicle.

(64) In various embodiments, the vehicle then proceeds to step **408** and determines whether the unknown device has breached the vehicle environment or is detected as being inside the vehicle. For example, if the relative location of the device is detected as being within the frame of the

vehicle, it may mean that the device is located either inside the vehicle, on top of the vehicle, or underneath the vehicle. If such is not true, then process **400** may proceed to step **410** and continue to monitor the location of the device for a period of time at step **410**. If, within this period of time, the device departs from the proximity threshold, then it may be deemed to be non-threatening. For example, if a stranger walks up to the vehicle to briefly admire or inspect the vehicle and then continues along his or her way, then the process will return to step **402** and will continue to monitor for other devices using the UWB anchors.

(65) However, if at step **410** the unknown device has not departed significantly in range, then the vehicle will return to step **406** and continue to localize the unknown device and continue to monitor the device's location relative to the vehicle. This is to ensure that the unknown individual is not merely lingering around the vehicle for an opportune moment to perpetrate a theft.

(66) Returning to step **408**, if the unknown device is determined to breach the vehicle environment or is detected as being within the frame of the vehicle (e.g., the device is either above, inside, or below the vehicle), then protection measures may be taken. In some embodiments, optional step **412** involves the use of other sensors to validate an intrusion. The other sensors may include, for example, activating microphones to detect loud noises or sawing, video cameras to detect whether there are persons above, inside, or beneath the car, motion sensors to detect whether certain vehicle components are being subjected to excessive or unexpected physical forces, and so on and so forth.

(67) Regardless, assuming that a sufficient threat level has been determined, the process **400** proceeds to step **414** and various protective measures may be taken. For example, a panic alarm may be initiated, the vehicle owner may be notified of a potential threat, and so on and so forth. In various embodiments, different alarms may be signaled based on different disturbance levels. For example, there may be an initial assessment period in which no alarms are triggered while the exact nature of the threat is being assessed. A disturbance level 1 may involve a single chirp and may be the first level of escalation. If this is not enough to cause the unknown device and unknown individual to leave the area, then a disturbance level 2 alarm may be performed with three chirps, and so on and so forth. A disturbance level 3 alarm may be a 23% duty cycle alarm (e.g., 220 ms alarm followed by 750 ms pause) and, if needed, followed by a disturbance level 4 alarm with a 50% duty cycle alarm (e.g., 220 ms alarm followed by 220 ms pause).

(68) The operations and processes described and shown above may be carried out or performed in any suitable order as desired in various implementations. Additionally, in certain implementations, at least a portion of the operations may be carried out in parallel. Furthermore, in certain implementations, less than or more than the operations described may be performed.

(69) The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any embodiment described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other embodiments. The terms “computing device,” “user device,” “communication station,” “station,” “handheld device,” “mobile device,” “wireless device” and “user equipment” (UE) as used herein refers to a wireless communication device such as a cable box, a wearable smart device, cellular telephone, a smartphone, a tablet, a netbook, a wireless terminal, a laptop computer, a femtocell, a high data rate (HDR) subscriber station, an access point, a printer, a point of sale device, an access terminal, or other personal communication system (PCS) device. The device may be either mobile or stationary.

(70) As used within this document, the term “communicate” is intended to include transmitting, or receiving, or both transmitting and receiving. This may be particularly useful in claims when describing the organization of data that is being transmitted by one device and received by another, but only the functionality of one of those devices is required to infringe the claim. Similarly, the bidirectional exchange of data between two devices (both devices transmit and receive during the exchange) may be described as “communicating,” when only the functionality of one of those devices is being claimed. The term “communicating” as used herein with respect to a wireless communication signal includes transmitting the wireless communication signal and/or receiving the

wireless communication signal. For example, a wireless communication unit, which is capable of communicating a wireless communication signal, may include a wireless transmitter to transmit the wireless communication signal to at least one other wireless communication unit, and/or a wireless communication receiver to receive the wireless communication signal from at least one other wireless communication unit.

(71) As used herein, unless otherwise specified, the use of the ordinal adjectives “first,” “second,” “third,” etc., to describe a common object, merely indicates that different instances of like objects are being referred to and are not intended to imply that the objects so described must be in a given sequence, either temporally, spatially, in ranking, or in any other manner.

(72) Some embodiments may be used in conjunction with various devices and systems, for example, a wearable smart device, a personal computer (PC), a desktop computer, a mobile computer, a laptop computer, a notebook computer, a tablet computer, a server computer, a handheld computer, a handheld device, a personal digital assistant (PDA) device, a handheld PDA device, an on-board device, an off-board device, a hybrid device, a vehicular device, a non-vehicular device, a mobile or portable device, a consumer device, a non-mobile or non-portable device, a wireless communication station, a wireless communication device, a wireless access point (AP), a wired or wireless router, a wired or wireless modem, a video device, an audio device, an audio-video (A/V) device, a wired or wireless network, a wireless area network, a wireless video area network (WVAN), a local area network (LAN), a wireless LAN (WLAN), a personal area network (PAN), a wireless PAN (WPAN), and the like.

(73) Some embodiments may be used in conjunction with one way and/or two-way radio communication systems, cellular radio-telephone communication systems, a mobile phone, a cellular telephone, a wireless telephone, a personal communication system (PCS) device, a PDA device which incorporates a wireless communication device, a mobile or portable global positioning system (GPS) device, a device which incorporates a GPS receiver or transceiver or chip, a device which incorporates an RFID element or chip, a multiple input multiple output (MIMO) transceiver or device, a single input multiple output (SIMO) transceiver or device, a multiple input single output (MISO) transceiver or device, a device having one or more internal antennas and/or external antennas, digital video broadcast (DVB) devices or systems, multi-standard radio devices or systems, a wired or wireless handheld device, e.g., a smartphone, a wireless application protocol (WAP) device, or the like.

(74) Some embodiments may be used in conjunction with one or more types of wireless communication signals and/or systems following one or more wireless communication protocols, for example, DOCSIS, radio frequency (RF), infrared (IR), frequency-division multiplexing (FDM), orthogonal FDM (OFDM), time-division multiplexing (TDM), time-division multiple access (TDMA), extended TDMA (E-TDMA), general packet radio service (GPRS), extended GPRS, code-division multiple access (CDMA), wideband CDMA (WCDMA), CDMA 2000, single-carrier CDMA, multi-carrier CDMA, multi-carrier modulation (MDM), discrete multi-tone (DMT), Bluetooth®, global positioning system (GPS), Wi-Fi, Wi-Max, ZigBee, ultra-wideband (UWB), global system for mobile communications (GSM), 2G, 2.5G, 3G, 3.5G, 4G, fifth generation (5G) mobile networks, 3GPP, long term evolution (LTE), LTE advanced, enhanced data rates for GSM Evolution (EDGE), or the like. Other embodiments may be used in various other devices, systems, and/or networks.

(75) Embodiments according to the disclosure are in particular disclosed in the attached claims directed to a method, a storage medium, a device and a computer program product, wherein any feature mentioned in one claim category, e.g., method, can be claimed in another claim category, e.g., system, as well. The dependencies or references back in the attached claims are chosen for formal reasons only. However, any subject matter resulting from a deliberate reference back to any previous claims (in particular multiple dependencies) can be claimed as well, so that any combination of claims and the features thereof are disclosed and can be claimed regardless of the

dependencies chosen in the attached claims. The subject-matter which can be claimed comprises not only the combinations of features as set out in the attached claims but also any other combination of features in the claims, wherein each feature mentioned in the claims can be combined with any other feature or combination of other features in the claims. Furthermore, any of the embodiments and features described or depicted herein can be claimed in a separate claim and/or in any combination with any embodiment or feature described or depicted herein or with any of the features of the attached claims.

(76) The foregoing description of one or more implementations provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

(77) Certain aspects of the disclosure are described above with reference to block and flow diagrams of systems, methods, apparatuses, and/or computer program products according to various implementations. It will be understood that one or more blocks of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and the flow diagrams, respectively, may be implemented by computer-executable program instructions. Likewise, some blocks of the block diagrams and flow diagrams may not necessarily need to be performed in the order presented, or may not necessarily need to be performed at all, according to some implementations.

(78) These computer-executable program instructions may be loaded onto a special-purpose computer or other particular machine, a processor, or other programmable data processing apparatus to produce a particular machine, such that the instructions that execute on the computer, processor, or other programmable data processing apparatus create means for implementing one or more functions specified in the flow diagram block or blocks. These computer program instructions may also be stored in a computer-readable storage media or memory that may direct a computer or other programmable data processing apparatus to function in a particular manner, such that the instructions stored in the computer-readable storage media produce an article of manufacture including instruction means that implement one or more functions specified in the flow diagram block or blocks. As an example, certain implementations may provide for a computer program product, comprising a computer-readable storage medium having a computer-readable program code or program instructions implemented therein, said computer-readable program code adapted to be executed to implement one or more functions specified in the flow diagram block or blocks. The computer program instructions may also be loaded onto a computer or other programmable data processing apparatus to cause a series of operational elements or steps to be performed on the computer or other programmable apparatus to produce a computer-implemented process such that the instructions that execute on the computer or other programmable apparatus provide elements or steps for implementing the functions specified in the flow diagram block or blocks.

(79) Accordingly, blocks of the block diagrams and flow diagrams support combinations of means for performing the specified functions, combinations of elements or steps for performing the specified functions and program instruction means for performing the specified functions. It will also be understood that each block of the block diagrams and flow diagrams, and combinations of blocks in the block diagrams and flow diagrams, may be implemented by special-purpose, hardware-based computer systems that perform the specified functions, elements or steps, or combinations of special-purpose hardware and computer instructions.

(80) Implementations of the systems, apparatuses, devices, and methods disclosed herein may comprise or utilize a special purpose or general-purpose computer including computer hardware, such as, for example, one or more processors and system memory, as discussed herein. Computer-executable instructions comprise, for example, instructions and data which, when executed at a processor, cause a general-purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions. An implementation of the

devices, systems and methods disclosed herein may communicate over a computer network. A “network” is defined as one or more data links that enable the transport of electronic data between computer systems and/or modules and/or other electronic devices.

(81) Although the subject matter has been described in language specific to structural features and/or methodological acts, it is to be understood that the subject matter defined in the appended claims may not necessarily be limited to the described features or acts described above. Rather, the described features and acts are disclosed as example forms of implementing the claims.

(82) While various embodiments of the present disclosure have been described above, it should be understood that they have been presented by way of example only, and not limitation. It will be apparent to persons skilled in the relevant art that various changes in form and detail can be made therein without departing from the spirit and scope of the present disclosure. Thus, the breadth and scope of the present disclosure should not be limited by any of the above-described exemplary embodiments but should be defined only in accordance with the following claims and their equivalents. The foregoing description has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the present disclosure to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. Further, it should be noted that any or all of the aforementioned alternate implementations may be used in any combination desired to form additional hybrid implementations of the present disclosure. For example, any of the functionality described with respect to a particular device or component may be performed by another device or component. Conditional language, such as, among others, “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments could include, while other embodiments may not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments.

Claims

1. A vehicle, comprising: a plurality of Ultra-Wideband (UWB) primary anchors; one or more second sensors, wherein the one or more second sensors are different than the UWB anchors; and one or more processors and memory storing executable instructions that, as a result of execution by the one or more processors, causes the vehicle to: determine, using at least one UWB anchor of the plurality of UWB primary anchors, presence of an unknown device; determine, using the at least one UWB anchor, a distance of the unknown device to the vehicle; determine, based on the distance, that the unknown device has crossed a proximity threshold of the vehicle; determine, using the plurality of UWB primary anchors, a location of the unknown device; determine, based on the location of the unknown device, that the unknown device is within a boundary defined by the vehicle's frame; and responsive to the determination that the unknown device is within the boundary defined by the vehicle's frame, activate the one or more second sensors; and determine, using the one or more second sensors, that the unknown device is located underneath the vehicle and an unknown user associated with the unknown device is preparing for or is currently engaging in a theft of a vehicle component from underneath the vehicle.
2. The vehicle of claim 1, wherein the executable instructions, as a result of execution by the one or more processors, further causes the vehicle to: determine, based on additional data collected by the one or more second sensors of the vehicle, a threat level; and determine one or more protection steps based at least in part on the threat level.
3. The vehicle of claim 2, wherein the one or more second sensors comprise audio and visual sensors.
4. The vehicle of claim 2, wherein the one or more second sensors comprise a camera underneath the vehicle.

5. The vehicle of claim 2, wherein the one or more protection steps comprises: activation of a first alarm; and activation of a second alarm, responsive to a determination that the unknown device is still within the proximity threshold after the activation of the first alarm.
6. The vehicle of claim 1, wherein the proximity threshold is determined based at least in part on time of day.
7. The vehicle of claim 1, wherein the executable instructions, as a result of execution by the one or more processors, further causes the vehicle to: determine that a known device associated with the vehicle is not within proximity to the vehicle.
8. A method, comprising: determining, using at least one Ultra-Wideband (UWB) anchor of a vehicle, presence of an unknown device; determining, using the at least one UWB anchor, a distance of the unknown device to the vehicle; determining, based on the distance, that the unknown device has crossed a proximity threshold of the vehicle; determining, using a plurality of UWB anchors of the vehicle, a location of the unknown device; determining, based on the location of the unknown device, that the unknown device is within a boundary defined by the vehicle's frame; responsive to the determination that the unknown device is within the boundary defined by the vehicle's frame, activating one or more second sensors, wherein the one or more second sensors are different than the UWB anchors; and determining, using the one or more second sensors, that the unknown device is located underneath the vehicle and an unknown user associated with the unknown device is preparing for or is currently engaging in a theft of a vehicle component from underneath the vehicle.
9. The method of claim 8, further comprising: determining, based on additional data collected by the one or more second sensors of the vehicle, a threat level; and determining one or more protection steps based at least in part on the threat level.
10. The method of claim 9, wherein the one or more second sensors comprise audio and visual sensors.
11. The method of claim 9, wherein the one or more second sensors comprise a camera underneath the vehicle.
12. The method of claim 9, wherein performing the one or more protection steps comprises: activating a first alarm; determining that the unknown device is still within the proximity threshold after activating the first alarm; and activating a second alarm.
13. The method of claim 8, wherein the proximity threshold is determined based at least in part on time of day.
14. The method of claim 8, further comprising determining that a known device associated with the vehicle is not within proximity to the vehicle.
15. A non-transitory computer-readable storage medium storing executable instructions that, as a result of being executed by one or more processors of an on-board system of a vehicle, cause the vehicle to at least: determine, using at least one Ultra-Wideband (UWB) primary anchor of the vehicle, presence of an unknown device; determine, using the at least one UWB anchor, a distance of the unknown device to the vehicle; determine, based on the distance, that the unknown device has crossed a proximity threshold of the vehicle; determine, using a plurality of UWB primary anchors of the vehicle, a location of the unknown device; determine, based on the location of the unknown device, that the unknown device is within a boundary defined by the vehicle's frame; responsive to the determination that the unknown device is within the boundary defined by the vehicle's frame, activate one or more second sensors, wherein the one or more second sensors are different than the UWB anchors; and determine, using the one or more second sensors, that the unknown device is located underneath the vehicle and an unknown user associated with the unknown device is preparing for or is currently engaging in a theft of a vehicle component from underneath the vehicle.
16. The non-transitory computer-readable storage medium of claim 15, wherein the executable instructions, as a result of execution by the one or more processors, further causes the vehicle to:

- determine, based on additional data collected by the one or more second sensors of the vehicle, a threat level; and determine one or more protection steps based at least in part on the threat level.
17. The non-transitory computer-readable storage medium of claim 16, wherein the one or more second sensors comprise audio and visual sensors.
18. The non-transitory computer-readable storage medium of claim 16, wherein the one or more second sensors comprise a camera underneath the vehicle.
19. The non-transitory computer-readable storage medium of claim 16, wherein the one or more protection steps comprises: activation of a first alarm; and activation of a second alarm, responsive to a determination that the unknown device is still within the proximity threshold after the activation of the first alarm.
20. The non-transitory computer-readable storage medium of claim 15, wherein the proximity threshold is determined based at least in part on time of day.
-